

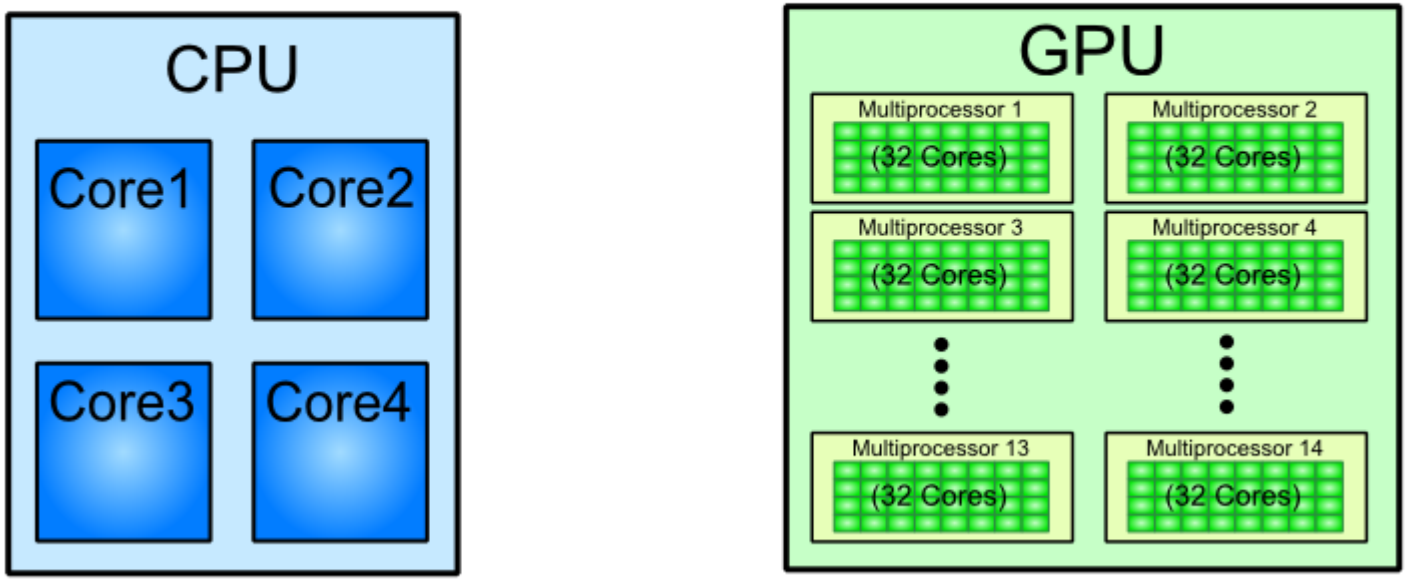
Computación acelerada (GPU/TPU)

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SCC-ETSI Informática-UNED

GPU

- *Fuente:* https://www.maketecheasier.com/difference-between-cpu-and-gpu/

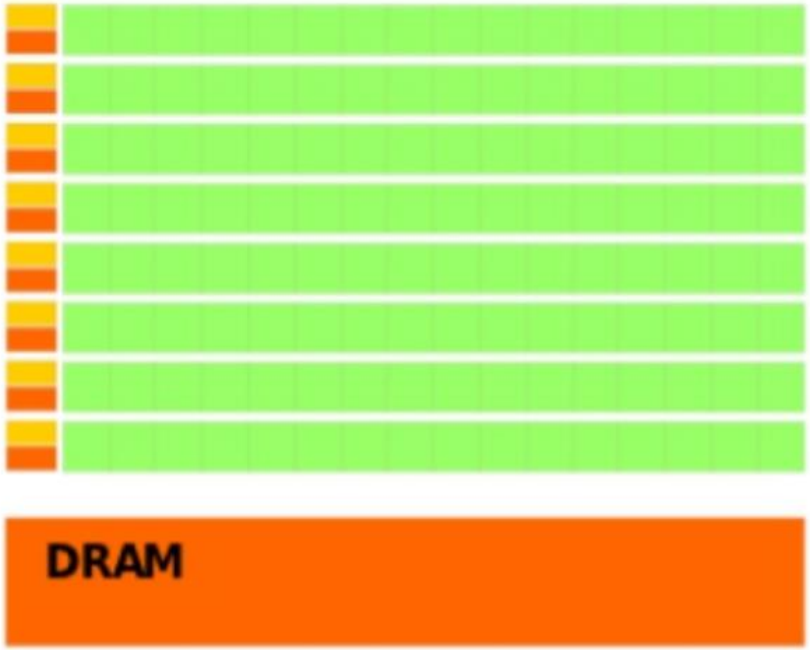
CPU/GPU Architecture Comparison



CPU VERSUS GPU

- El hardware de la GPU tiene muchas más unidades lógicas que una CPU
  - Mejor capacidad para procesar en paralelo operaciones aritméticas (multiplicación de matrices, etc.)





Fuente: https://www.slideshare.net/VishalSingh405/cpu-vs-gpu-presentation-54700475

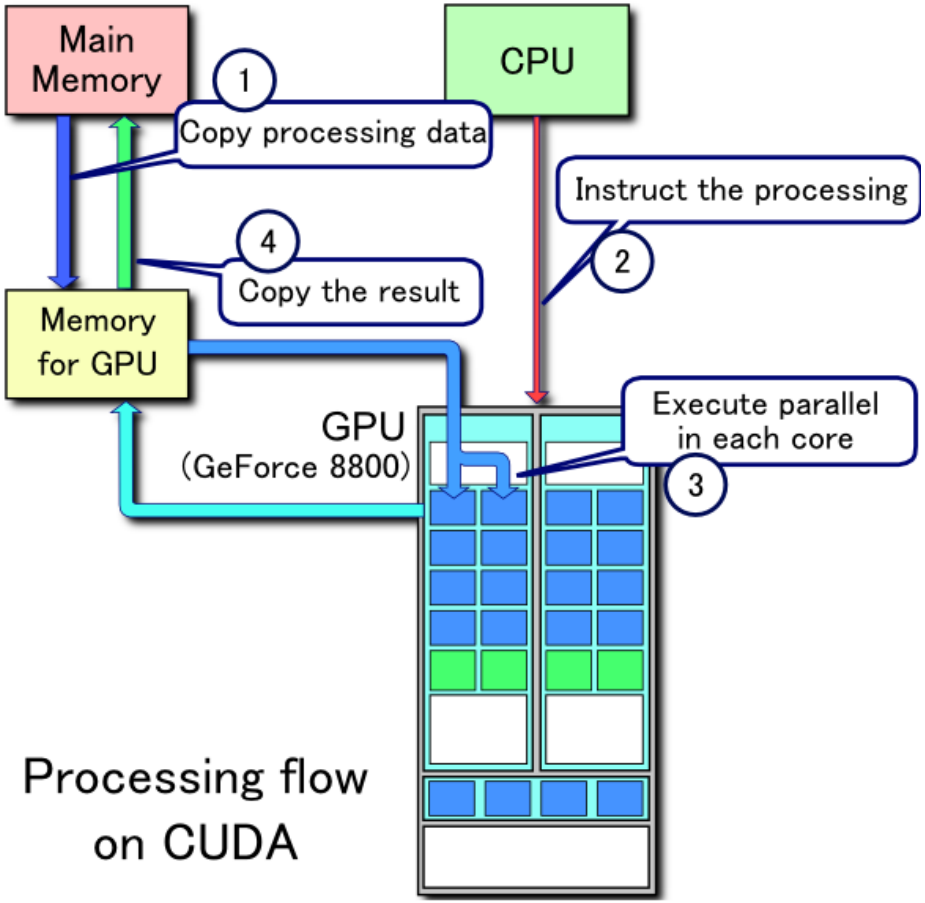
GPU GANANCIA EN VELOCIDAD: SHORTEST PATH

Total Vertices	Size of Adjacency Matrix	CPU time(s)	GPU time(s)	Speedup
1000	1,000,000	3.9s	0.103s	37.86x
2000	4,000,000	30.90s	0.698s	44.34x
4000	16,000,000	244.22s	5.09s	47.98x
8000	64,000,000	1941.0s	39.1s	49.64x
10000	100,000,000	3778.1s	77.8s	48.56
11111	123,454,321	5179.2s	108.01s	47.95

TARJETAS Y PROVEEDORES

- NVIDIA
  - Compute Unified Device Architecture (CUDA)
  - NVLink, GPUDirect
  - Evoluciones: … Pascal, Volta y Turing
    - https://www.nvidia.com/es-es/about-nvidia/corporate-timeline/
- AMD
  - Heterogeneous System Architecture (HSA)
    - Multiplataforma
    - ROCm / OpenCL
  - Arquitecturas Vega GCN
  - Tarjetas Radeon (Radeon, Radeon Pro e Instinct)

PLATAFORMA CUDA (1/2)



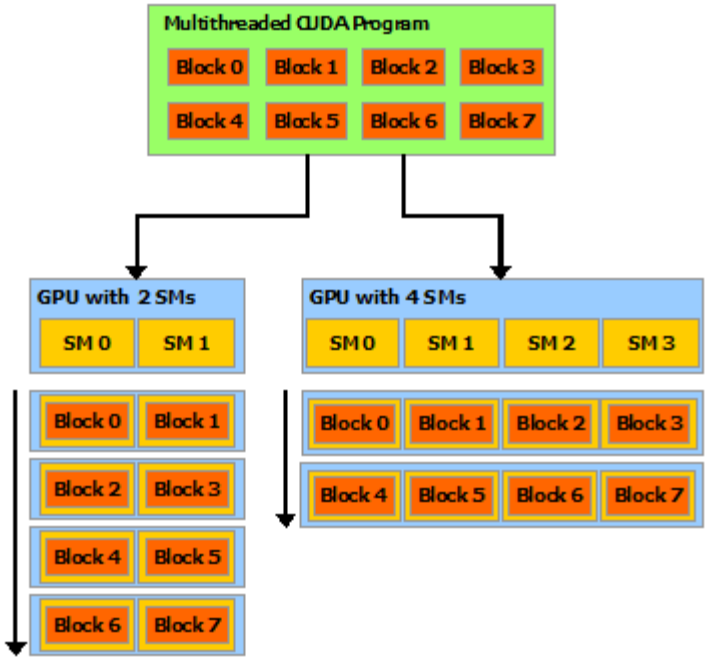


Figura: https://docs.nvidia.com/cuda/cuda-c-programming-guide/index.html

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PLATAFORMA CUDA (2/2)

GPU Computing Applications						
Libraries and Middleware						
cuDNN TensorRT	cuFFT, cuBLAS, cuRAND, cuSPARSE	CULA MAGMA	Thrust NPP	VSIPL, SVM, OpenCurrent	PhysX, OptiX, iRay	MATLAB Mathematica
Programming Languages						
C	C++	Fortran	Java, Python, Wrappers	DirectCompute	Directives (e.g., OpenACC)	
CUDA-enabled NVIDIA GPUs						
Turing Architecture (Compute capabilities 7.x)	DRIVE/JETSON AGX Xavier	GeForce 2000 Series	Quadro RTX Series	Tesla T Series		
Volta Architecture (Compute capabilities 7.x)	DRIVE/JETSON AGX Xavier			Tesla V Series		
Pascal Architecture (Compute capabilities 6.x)	Tegra X2	GeForce 1000 Series	Quadro P Series	Tesla P Series		
Maxwell Architecture (Compute capabilities 5.x)	Tegra X1	GeForce 900 Series	Quadro M Series	Tesla M Series		
Kepler Architecture (Compute capabilities 3.x)	Tegra K1	GeForce 700 Series GeForce 600 Series	Quadro K Series	Tesla K Series		
	EMBEDDED	CONSUMER DESKTOP, LAPTOP	PROFESSIONAL WORKSTATION	DATA CENTER		

ARQUITECTURA TURING/VOLTA

	GeForce Titan V	Quadro RTX 8000	Tesla V100
Microarchitecture	Volta	Turing	Volta
Memory size	12 GB	48 GB	16 GB
Memory type	HBM2	GDDR6	HBM2
Memory bus	3,072-bit	384-bit	4,096-bit
Bandwidth	652.8 GB/s	672.0 GB/s	897.0 GB/s
NVLink support	No	Yes	Yes
GPUDirect support	Limited	Full	Full

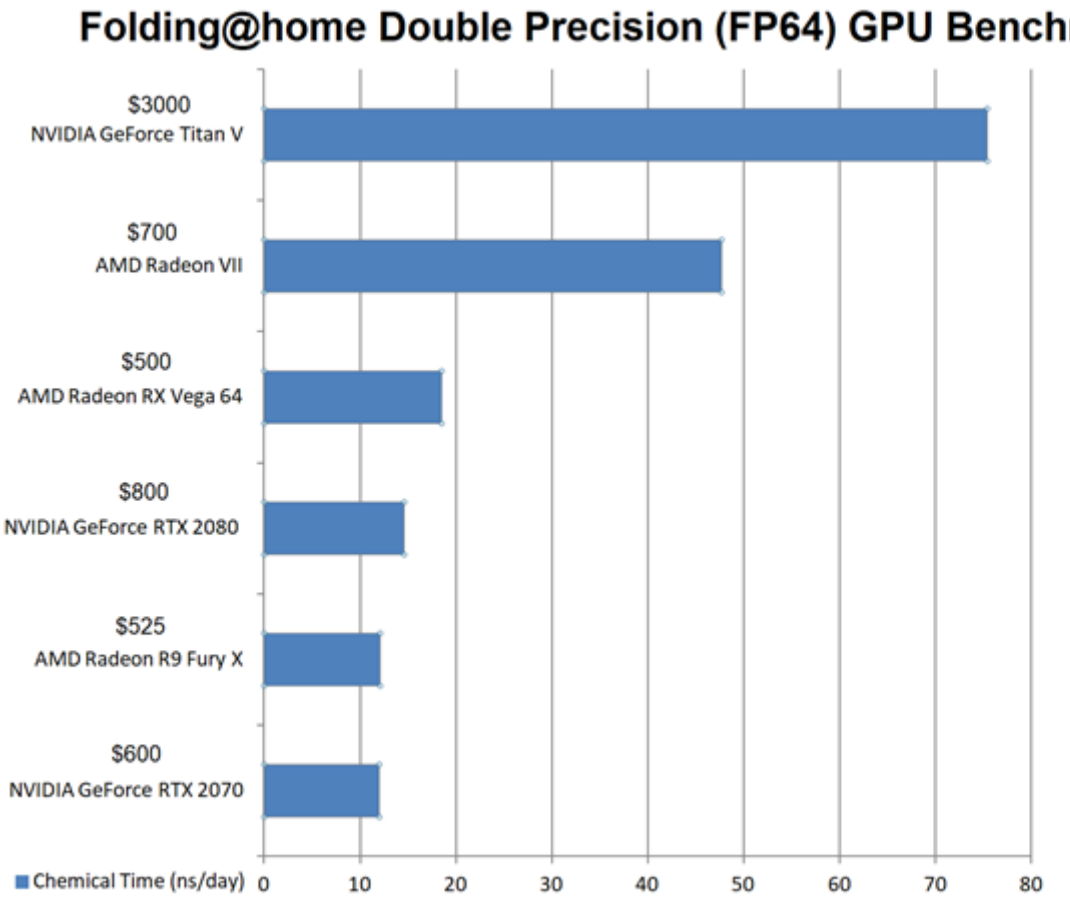
	GeForce Titan V	Quadro RTX 8000	Tesla V100
Pixel Rate	~139.7 GPixel/s	~169.9 GPixel/s	~176.6 GPixel/s
Texture rate	~465.6 GTexel/s	~509.8 GTexel/s	~441.6 GTexel/s
FP16 (half) performance	~27.5 TFLOPS	32.6 TFLOPS	~31.4 TFLOPS
FP32 (float) performance	~14.8 TFLOPS	~16.3 TFLOPs	~14.1 TFLOPS
FP64 (double) performance	Up to 6.875 TFLOPS	~0.5 TFLOPS	~7.8 TFLOPS

Deep Learning ➔

	GeForce Titan V	Quadro RTX 8000	Tesla V100
TensorTFLOPS	110	130.5	125

Fuente: Hands-On GPU Computing with Python

COMPARATIVA/BENCHMARK



GPU	Titan V	Radeon VII
Double-precision performance	6.9 TFLOPS FP64	3.46 TFLOPS FP64
Memory	12 GB HBM2	16 GB HBM2
Bandwidth	652.8 GB/s	1 TB/s
Price	$3,000	$700

Fuente: Hands-On GPU Computing with Python

BENCHMARKS EN TENSORFLOW

- https://github.com/tensorflow/models/tree/master/official/benchmark
- https://dawn.cs.stanford.edu/benchmark/

  - Image Classification (ImageNet)
  - Image Classification (CIFAR10)
    - Training Time
    - Training Cost
    - Inference Latency
    - Inference Cost
  - Question Answering (SQuAD)

Training Time

All Submissions

Objective: Time taken to train an image classification model to a top-5 validation accuracy of 93% or greater on ImageNet.

Rank	Time to 93% Accuracy	Model	Hardware	Framework
1 Mar 2020	0:02:38	ResNet50-v1.5 *Apsara AI Acceleration(AIACC) team in Alibaba Cloud* source	16 ecs.gn6e-c12g1.24xlarge (AlibabaCloud)	AIACC-Training 1.3 + Tensorflow 2.1
2 May 2019	0:02:43	ResNet-50 *ModelArts Service of Huawei Cloud* source	16 nodes with InfiniBand (8*V100 with NVLink for each node)	Moxing v1.13.0 + TensorFlow v1.13.1
3 Dec 2018	0:09:22	ResNet-50 *ModelArts Service of Huawei Cloud* source	16 * 8 * Tesla-V100(ModelArts Service)	Huawei Optimized MXNet
4 Sep 2018	0:18:06	ResNet-50 *fast.ai/DIUx (Yaroslav Bulatov, Andrew Shaw, Jeremy Howard)* source	16 p3.16xlarge (AWS)	PyTorch 0.4.1
5 Sep 2018	0:18:53	Resnet 50 *Andrew Shaw, Yaroslav Bulatov, Jeremy Howard* source	64 * V100 (8 machines - AWS p3.16xlarge)	ncluster / Pytorch 0.5.0a0+0e8088d

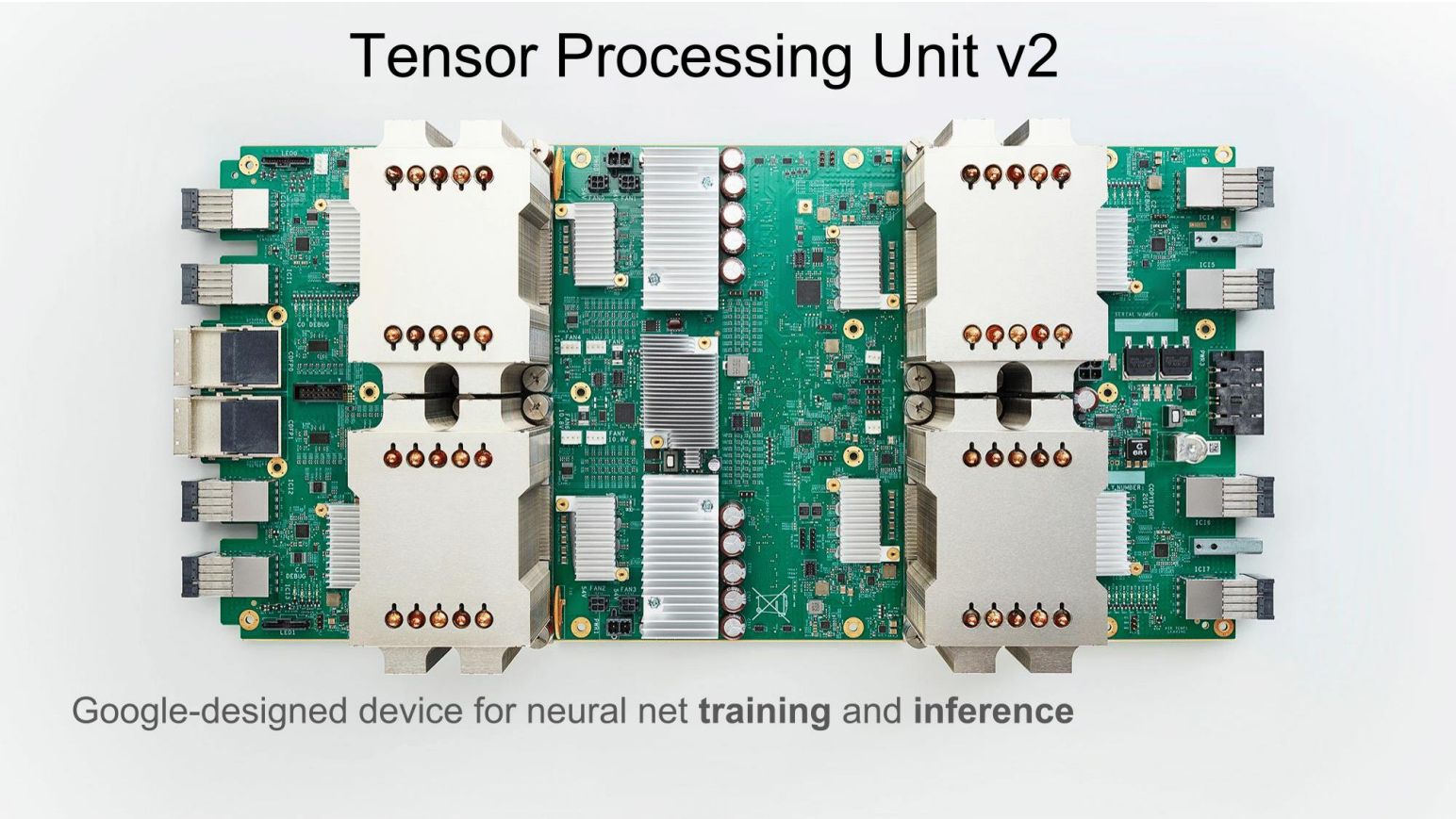
GPU Y CPU CON REDES NEURONALES

- https://blog.algorithmia.com/hardware-for-machine-learning/

- http://minimaxir.com/2017/07/cpu-or-gpu/
- https://medium.com/@andriylazorenko/tensorflow-performance-test-cpu-vs-gpu-79fcd39170c

- Selección de GPU
  - https://blog.slavv.com/picking-a-gpu-for-deep-learning-3d4795c273b9
  - https://www.sisoftware.co.uk/benchmarking-101/

TPU (TENSOR PROCESSON UNIT)



- https://cloud.google.com/tpu/docs/quickstart
- https://cloud.google.com/tpu/docs/tutorials

DISPONIBILIDAD EN LA NUBE

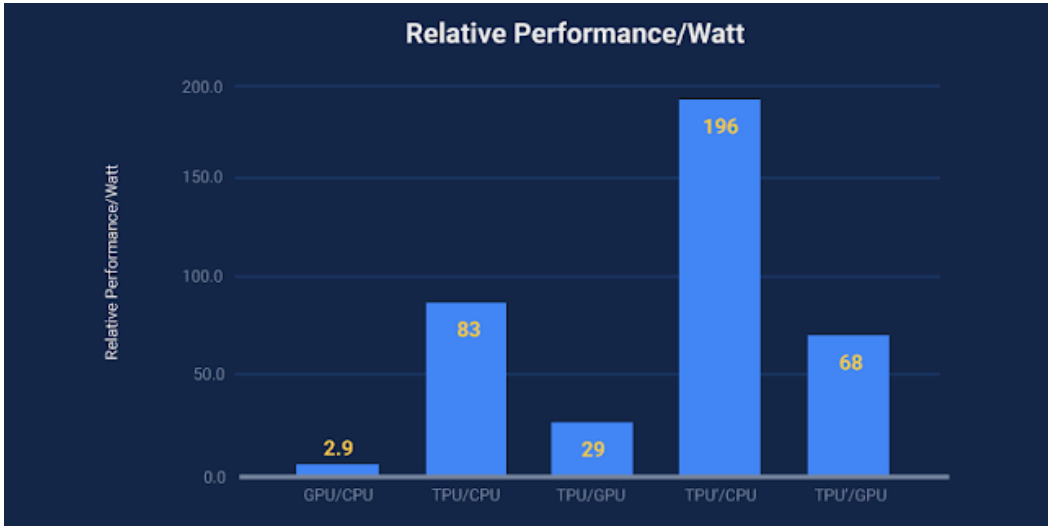
- https://cloudplatform.googleblog.com/2016/05/Google-supercharges-machine-learning-tasks-with-custom-chip.html



	nVidia k80	TPU
FLOPs	9 Teraflops	180 Teraflops

¿POR QUÉ TPUS? SEGÚN GOOGLE… (1/2)

- https://cloudplatform.googleblog.com/2017/04/quantifying-the-performance-of-the-TPU-our-first-machine-learning-chip.html
  - 30x más rápido que GPUs y CPUs.
  - Eficiencia energética (30x a 80x en las medidas de TOPS/Watt)
  - NN desarrolladas/ejecutadas/modeladas requieren muy pocas líneas de código (Tensoflow)



POR QUÉ TPUS? SEGÚN GOOGLE… (2/2)

- Menos dinero y menos tiempo

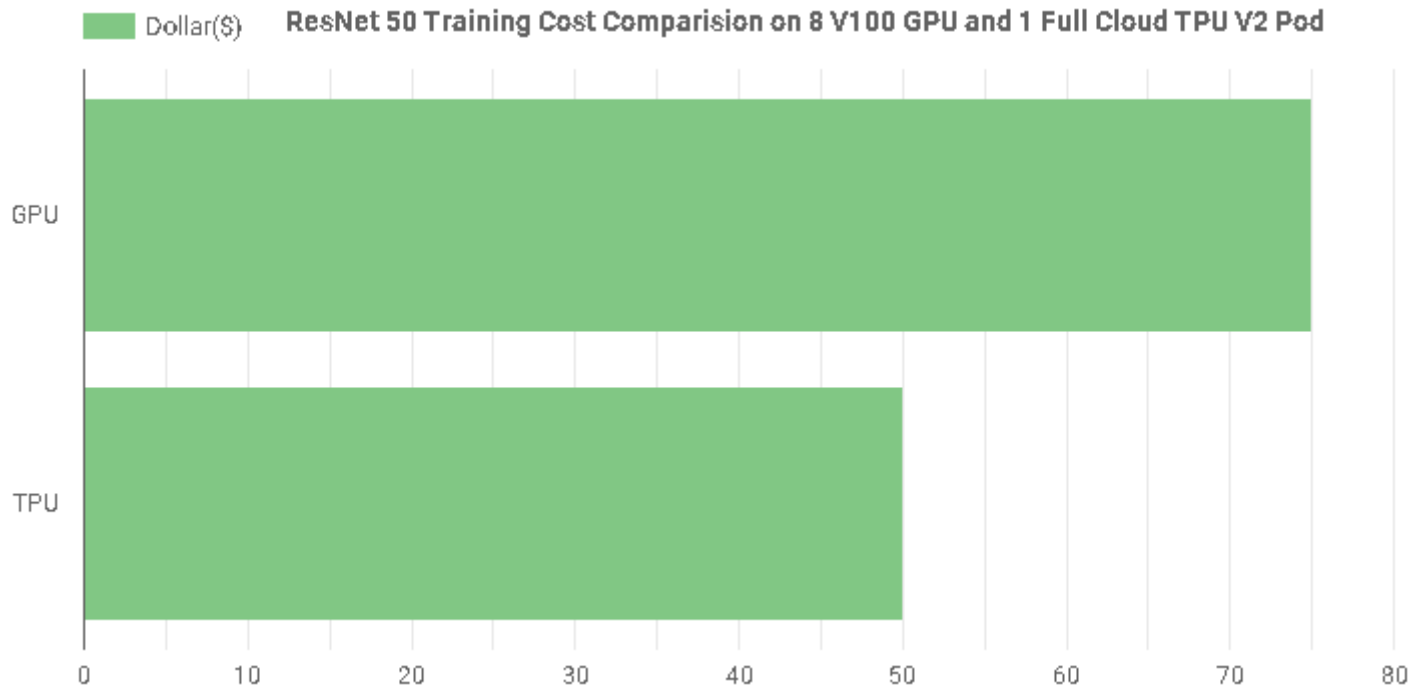


Figura del libro: Hands-On Artificial Intelligence on Google Cloud Platform
https://learning.oreilly.com/library/view/hands-on-artificial-intelligence/9781789538465/

CUANDO USAR TPU?

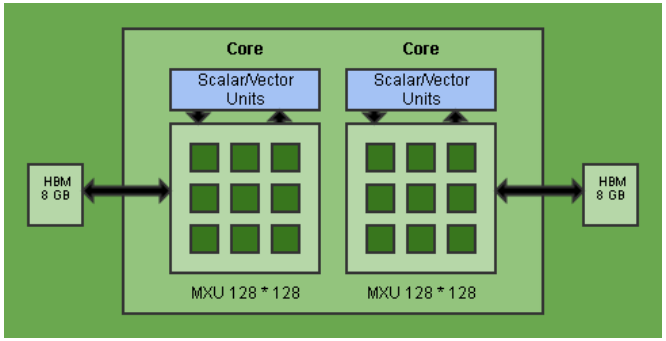
- Recomendaciones generales:

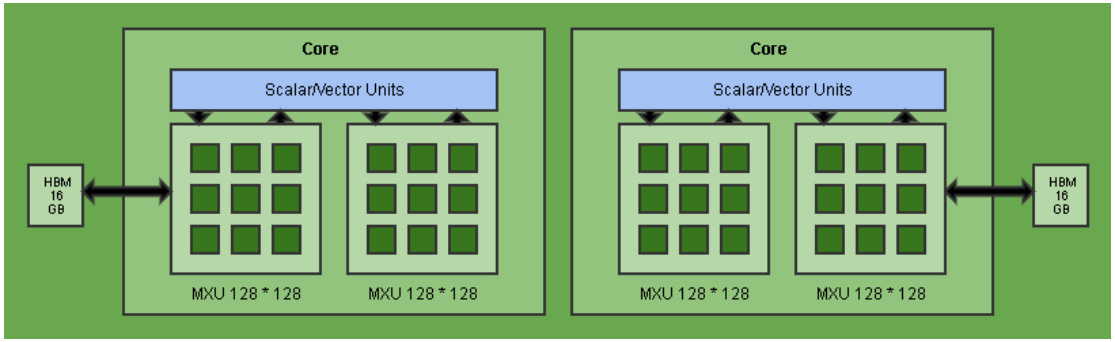
CPU	GPU	TPU
Quick prototyping that requires maximum flexibility	Models that are not written in TensorFlow	Models dominated by matrix computations
Simple models that do not take long to train	Models for which source does not exist or is too onerous to change	Models with no custom TensorFlow operations inside the main training loop
Small models with small effective batch sizes	Models with a significant number of custom TensorFlow operations that must run at least partially on CPUs	Models that train for weeks or months
Models that are dominated by custom TensorFlow operations written in C++	Models with TensorFlow ops that are not available on Cloud TPU	Larger and very large models with very large effective batch sizes
Models that are limited by available I/O or the networking bandwidth of the host system	Medium-to-large models with larger effective batch sizes	Models suitable for linear algebra that does not require frequent branching

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VERSIONES

- Hay dos modelos:
  - V2 ➔HBM de 8 GB/TPU core. 1MXU (128*128) por core. TPU pod, hasta 512 cores (4TB de memoria)
  - V3 ➔ HBM de 16 GB/TPU core. 2 MXU (128*128) por core. TPU pod, hasta 2048 cores (32 TB de memoria)
- Una MXU puede ejcecutar hasta 16000 operaciones de multiplicación acumulativa (MAC) por ciclo (coma flotante de 32 bits)





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CONFIGURACIONES TPUS (2/2)
TPU SINGLE DEVICE

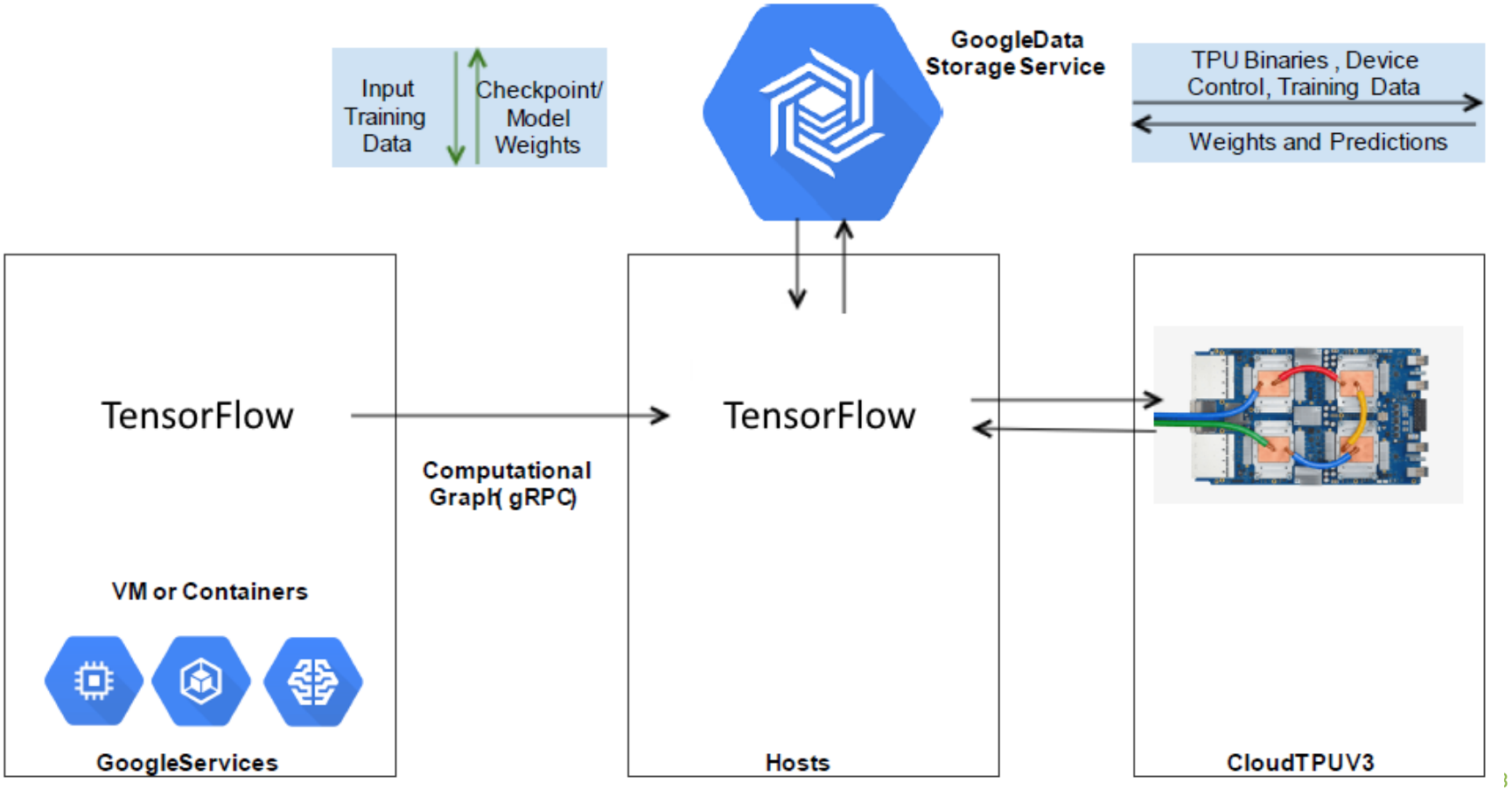


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CONFIGURACIONES TPUS (2/2)
TPU POD

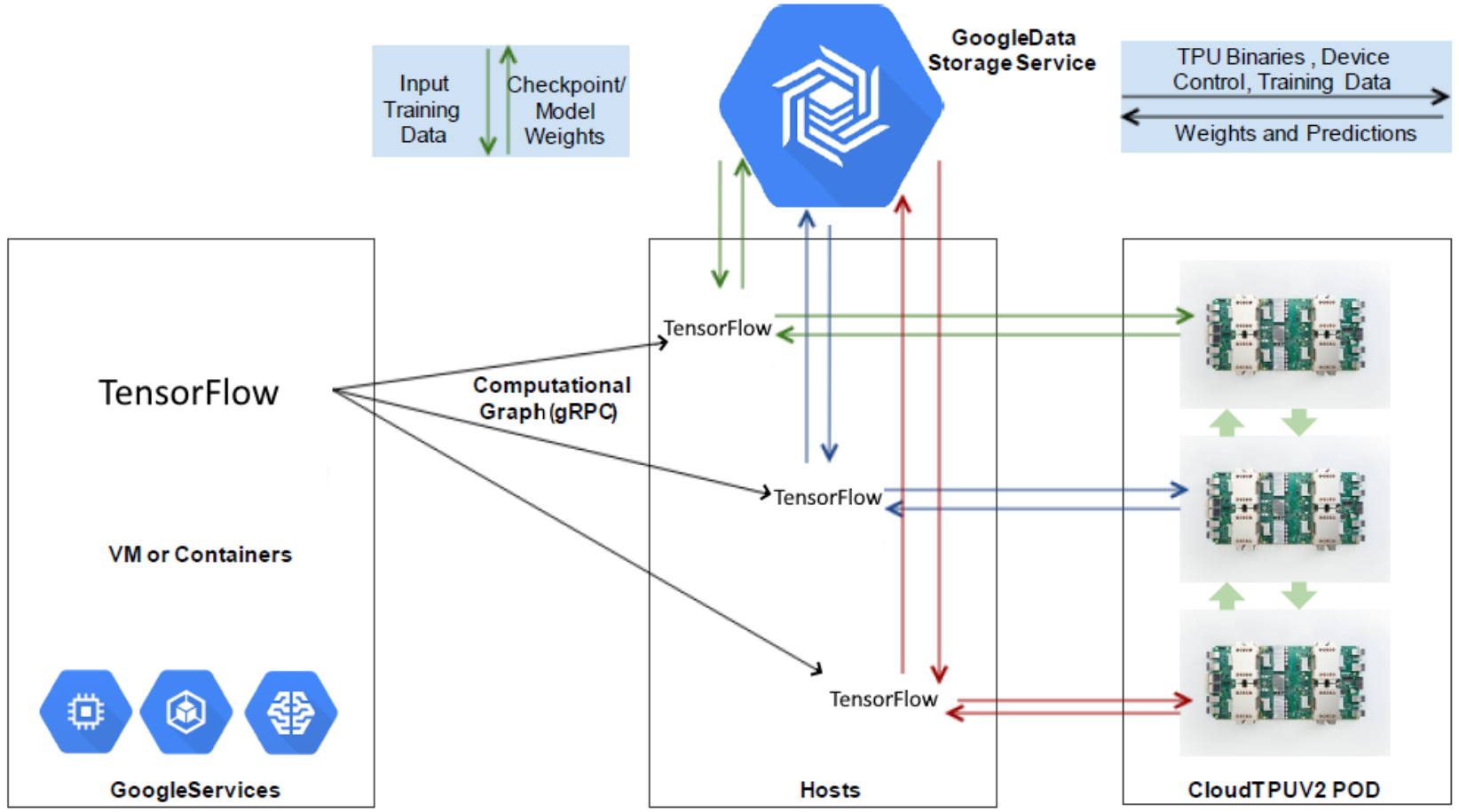


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FLUJO DE EJECUCIÓN CON TPUS

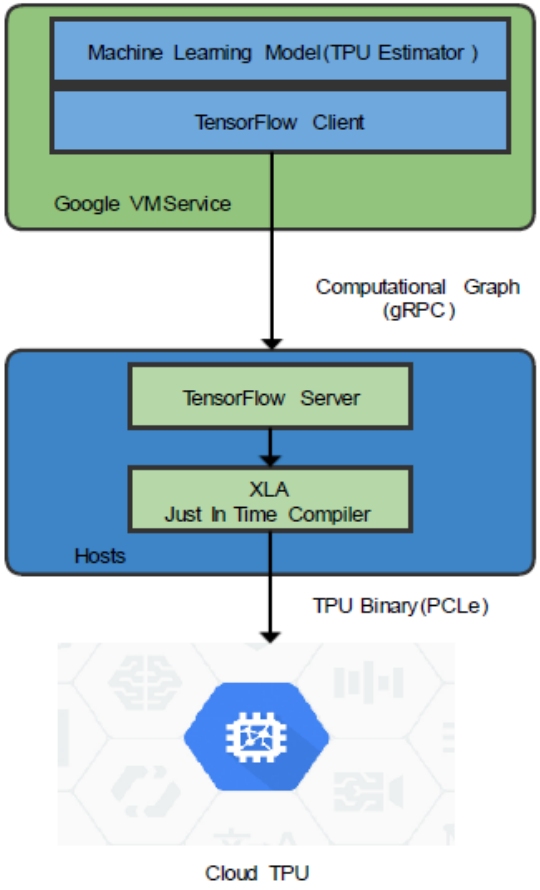


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GOOGLE COLABORATORY

- https://colab.research.google.com/
- EJERCICIO: Comparativa de rendimiento: CPU/GPU
  - https://github.com/moritzhambach/CPU-vs-GPU-benchmark-on-MNIST/blob/master/gputest.ipynb